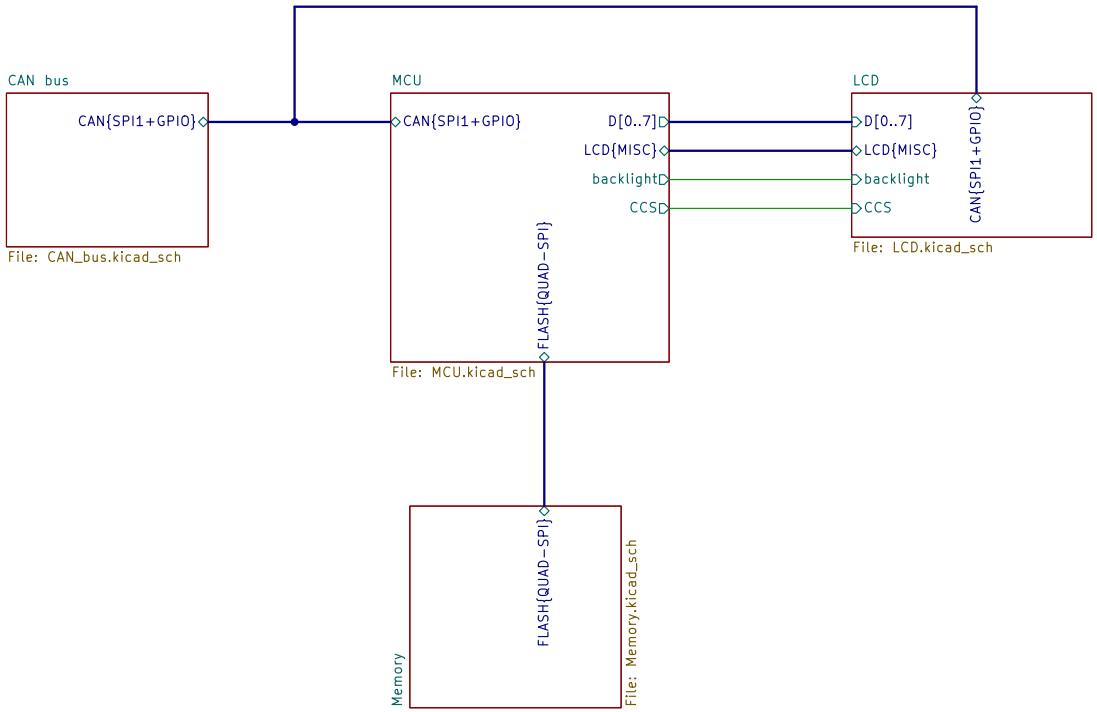


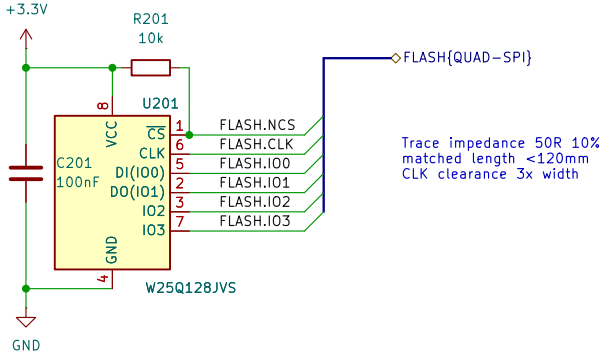
H101
MountingHole

H102
MountingHole

H103
MountingHole

Sheet: / File: BB_screen_main_V1.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 1/5

Quad-SPI FLASH NOR Memory

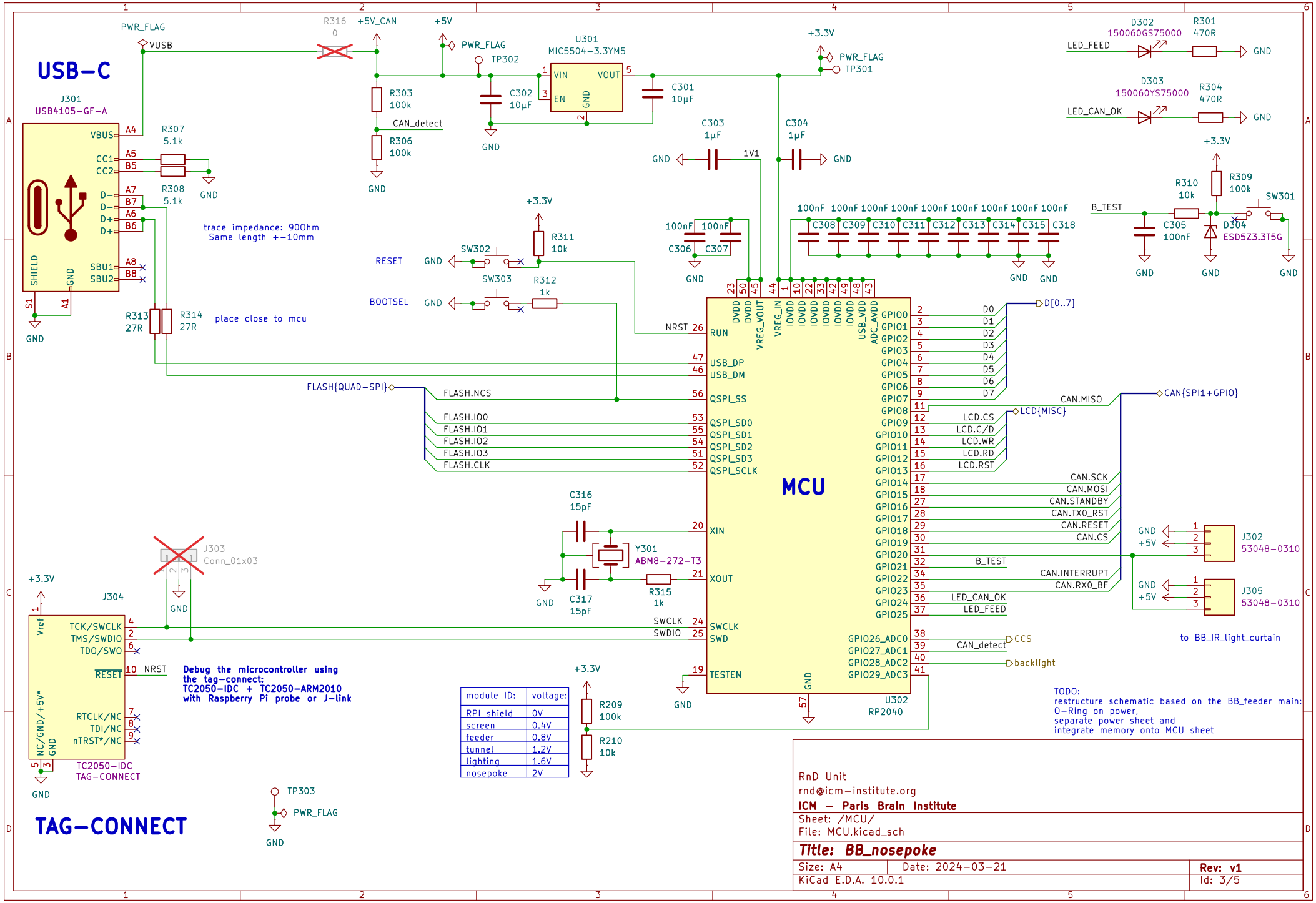


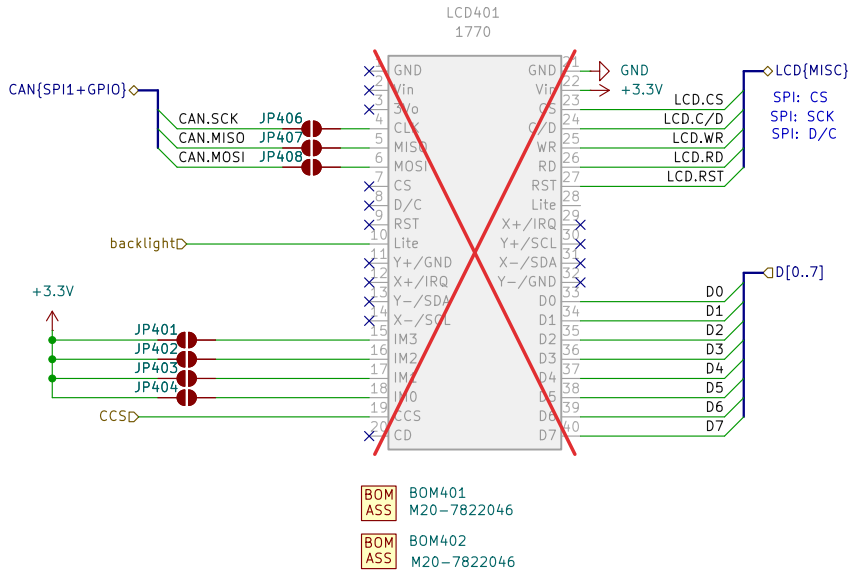
RnD Unit
rnd@icm-institute.org
ICM - Paris Brain Institute

Sheet: /Memory/
File: Memory.kicad_sch

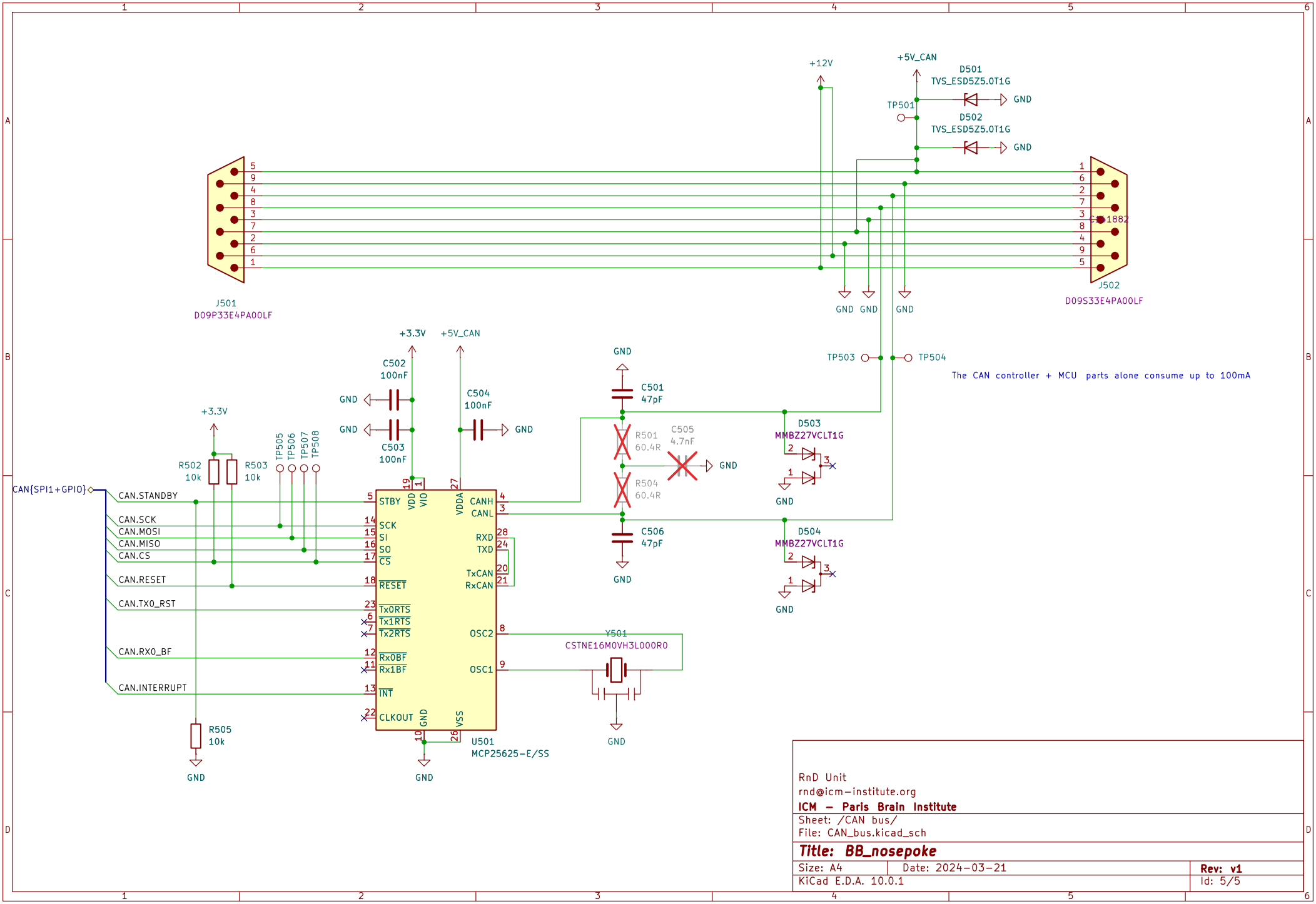
Title: BB_nosepoke

Size: A4	Date: 2024-03-21	**Rev: v1**
KiCad E.D.A. 10.0.1		Id: 2/5


USB-C
J301
USB4105-GF-A
VBUS
CC1
CC2
D-
D-
D+
D+
SHIELD
GND
SBU1
SBU2
S1
A1
PWR_FLAG
VUSB
R316
0
+5V_CAN
+5V
PWR_FLAG
TP302
U301
MIC5504-3.3YM5
VIN
VOUT
EN
GND
R303
100k
CAN_detect
R306
100k
C302
10µF
C301
10µF
+3.3V
PWR_FLAG
TP301
C303
1µF
1V1
C304
1µF
R307
5.1k
R308
5.1k
trace impedance: 90Ohm
Same length +-10mm
R313
27R
R314
27R
place close to mcu
+3.3V
R311
10k
RESET
SW302
BOOTSEL
SW303
R312
1k
100nF
C306
C307
100nF
C308
C309
C310
C311
C312
C313
C314
C315
C318
D302
150060GS75000
R301
470R
LED_FEED
D303
150060YS75000
R304
470R
LED_CAN_OK
R310
10k
R309
100k
SW301
B_TEST
C305
100nF
D304
ESD5Z3.3T5G
MCU
NRST
RUN
USB_DP
USB_DM
QSPI_SS
QSPI_SD0
QSPI_SD1
QSPI_SD2
QSPI_SD3
QSPI_SCLK
XIN
XOUT
SWCLK
SWD
TESTEN
FLASH{QUAD-SPI}
FLASH.NCS
FLASH.IO0
FLASH.IO1
FLASH.IO2
FLASH.IO3
FLASH.CLK
C316
15pF
C317
15pF
Y301
ABM8-272-T3
R315
1k
J303
Conn_01x03
J304
Vref
TCK/SWCLK
TMS/SWDIO
TDO/SWO
RESET
NRST
RTCLK/NC
TDI/NC
nTRST*/NC
NC/GND/+5V*
GND
TC2050-IDC
TAG-CONNECT
TAG-CONNECT
Debug the microcontroller using
the tag-connect:
TC2050-IDC + TC2050-ARM2010
with Raspberry Pi probe or J-link
TP303
PWR_FLAG
module ID:
voltage:
RPI shield
0V
screen
0.4V
feeder
0.8V
tunnel
1.2V
lighting
1.6V
nosepoke
2V
R209
100k
R210
10k
D[0..7]
D0
D1
D2
D3
D4
D5
D6
D7
LCD{MISC}
LCD.CS
LCD.C/D
LCD.WR
LCD.RD
LCD.RST
CAN.MISO
CAN{SPI1+GPIO}
CAN.SCK
CAN.MOSI
CAN.STANDBY
CAN.TX0_RST
CAN.RESET
CAN.CS
B_TEST
CAN.INTERRUPT
CAN.RX0_BF
LED_CAN_OK
LED_FEED
CCS
CAN_detect
backlight
GPIO26_ADC0
GPIO27_ADC1
GPIO28_ADC2
GPIO29_ADC3
U302
RP2040
J302
53048-0310
J305
53048-0310
to BB_IR_light_curtain
TODO:
restructure schematic based on the BB_feeder main:
O-Ring on power,
separate power sheet and
integrate memory onto MCU sheet
RnD Unit
rnd@icm-institute.org
ICM - Paris Brain Institute
Sheet: /MCU/
File: MCU.kicad_sch
Title: BB_nosepoke
Size: A4
Date: 2024-03-21
Rev: v1
KiCad E.D.A. 10.0.1
Id: 3/5



Sheet: /LCD/		
File: LCD.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.1		Id: 4/5


+12V
+5V_CAN
D501
TVS_ESD5Z5.0T1G
GND
TP501
D502
TVS_ESD5Z5.0T1G
GND
J501
D09P33E4PA00LF
J502
D09S33E4PA00LF
GND GND GND
TP503
TP504
The CAN controller + MCU parts alone consume up to 100mA
+3.3V
+5V_CAN
C502
100nF
GND
GND
C503
100nF
C504
100nF
GND
GND
C501
47pF
R501
60.4R
C505
4.7nF
GND
R504
60.4R
C506
47pF
GND
D503
MMBZ27VCLT1G
GND
D504
MMBZ27VCLT1G
GND
+3.3V
R502
10k
R503
10k
TP505
TP506
TP507
TP508
CAN{SPI1+GPIO}
CAN.STANDBY
CAN.SCK
CAN.MOSI
CAN.MISO
CAN.CS
CAN.RESET
CAN.TX0_RST
CAN.RX0_BF
CAN.INTERRUPT
STBY
SCK
SI
SO
CS
RESET
TX0RTS
TX1RTS
TX2RTS
RX0BF
RX1BF
INT
CLKOUT
VDD
VIO
VDDA
GND
VSS
CANH
CANL
RXD
TXD
TxCAN
RxCAN
OSC2
OSC1
Y501
CSTNE16M0VH3L000R0
GND
R505
10k
GND
GND
U501
MCP25625-E/SS
RnD Unit
rnd@icm-institute.org
ICM – Paris Brain Institute
Sheet: /CAN bus/
File: CAN_bus.kicad_sch
Title: BB_nosepoke
Size: A4
Date: 2024-03-21
Rev: v1
KiCad E.D.A. 10.0.1
Id: 5/5